Matched accuracy and cost controls for pre-export scaling

Research note accompanying the OMS manuscript

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Six known demand-inversion-free dispatch models, three previously used seeds, automatic/off presolve and four policies give 144 fixed assignments. This is a within-generator confirmation, not external validation. Targets are 0.01 MW and 100 m^3, checked exactly in original units after integer rounding on all 4,320 audited rows per saved point.

Presolve	Policy	Budgets	Max. v_V	$B > U$	Gap flags	Limits	Work	Total
Auto	L2	18/18	4.768e-07	5	3	0	1.59	1.944
Auto	Budget-L2	18/18	0.0001123	1	0	0	1.106	2.17
Auto	L2-tight	18/18	0	5	3	0	1.516	1.894
Auto	L2-NF3	18/18	0	0	0	0	3.063	2.295
Off	L2	15/18	4837	8	8	0	0.2904	1.343
Off	Budget-L2	18/18	1.478e-05	3	0	0	0.6023	1.795
Off	L2-tight	18/18	2.384e-07	8	8	0	0.2888	1.369
Off	L2-NF3	18/18	0	0	0	0	2.648	2.22

Every policy has 18 calls in each presolve configuration. Maximum activation violation is in m^3; Limits counts native time limits. Work is the geometric mean over all calls; Total is the median full fresh-process wall time in seconds. $B > U$ counts bounds contradicted by exact feasible witnesses of the actual export, allowing $\max(10^{-6}$,8 bound ulps). Gap flags count reported 1% completions satisfying $U < C - 0.01|C|$. Absence of a flag does not certify a bound or optimality.

Instance	L2	Budget-L2	L2-tight	L2-NF3
instance001	3/3	3/3	3/3	3/3
instance005	3/3	3/3	3/3	3/3
instance022	3/3	3/3	3/3	3/3
instance027	3/3	3/3	3/3	3/3
instance031	3/3	3/3	3/3	3/3
instance046	0/3	3/3	3/3	3/3

Per-model budget compliance, out of three seeds, with presolve disabled. Tight tolerances and NumericFocus are matched controls on the identical L2 export. All 144 terminal outcomes, including time limits, are retained. Objective values and paired Work ratios appear in the machine-readable reports; comparable primal budget compliance alone is not comparable optimization success.

Settings and cost scope. Gurobi 13.0.2, four threads, 120 seconds, 1% gap, ScaleFlag=-1, IntegralityFocus=0, seeds 11/23/47 and no MIP starts. L2/Budget-L2 use FeasibilityTol=10^{-6} and IntFeasTol=2^{-28}; L2-tight uses 10^{-9} for both; L2-NF3 adds NumericFocus=3 to the matched L2 tolerances. A fixed

permutation sets the order. The C++ generator constructs the model once per call; exports reproduce the prior model hashes. Only 1,440 activation factors per model differ for Budget-L2. Total includes startup, C++ generation/Base writing, metadata, factor selection, scaled export, native import/equality checks, optimization, extraction, verification and persistence. Stages are supplied; shared-machine wall times describe this prototype. Compilation and historical witness acquisition are offline costs; current witness revalidation is timed separately.

Scope. The experiment measures terminal delivery at the native stopping rule, not time to first verified incumbent or a full precision–cost frontier. Exact feasible witnesses refute specific bounds; they do not establish global optima. The reader-only replay checks twelve witnesses and every saved candidate without optimizer calls. The protocol, sources, logs, points, exact-unit checks and complete discussion are supplied alongside this note. OMS v1.3.8 and all previous closed research archives are preserved. A host reboot interrupted one additional attempt; its partial output is retained and that assignment alone was repeated, giving 145 native attempts and 144 terminal comparisons. The 37 completed calls before the reboot were kept; models, parameters and endpoints were unchanged.